

Waveform Characterization of Animal Contact, Tree Contact, and Lightning Induced Faults

Saurabh Kulkarni, Duehee Lee, Alicia J. Allen, Surya Santoso, Thomas A. Short

Abstract—In this paper signal processing tools are used to uncover common and unique characteristics of faults resulting from animal contacts, tree contacts and lightning. For each fault type a large number of voltage and current waveform data sets measured at monitoring stations on distribution systems are analyzed. The characteristics include but are not limited to the presence of impulse-like oscillations, the number of phases involved, the duration of fault event, the phase angle, the time of day, the spectral content in the time-frequency and time-scale domains, the rate of rise of voltage or current, and the arc voltage. An individual characteristic alone is insufficient to provide an estimate of the fault type. However, by combining common and unique characteristics extracted from a fault event, it may be possible to estimate the fault type accurately.

Index Terms—Diagnosis (fault), power distribution faults, power quality, power system monitoring, power system lightning effects, vegetation, animals, wavelet transforms

I. INTRODUCTION

POWER quality or PQ disturbances on distribution lines are caused by a number of factors [1] and can be internal or external. Internal events are primarily the result of equipment malfunction or failure, while external disturbances occur mainly due to animal and bird contacts, tree contacts, vehicle accidents as well as natural phenomena like lightning. The power quality monitoring stations that detect the deviations caused by these events store a snapshot of the voltage and current waveforms. This captured data possesses a large amount of information that can be analyzed to identify the root-cause of the disturbance.

A significant amount of work has been carried out in the areas of fault diagnosis, root-cause identification and automatic classification of power quality disturbance events. The use of wavelet domain analysis for fault cause identification is proposed in [2]. In addition, a number of artificial neural network based fault-cause identification methodologies have been traditionally discussed in literature [3]. For example, in [4] supervised clustering based neural networks are used to distinguish between faults and system conditions that appear like faults. A power distribution fault cause classifier developed in [5] makes use of logistic regression and artificial neural networks. A hybrid wavelet domain and artificial neural network based approach is

proposed in [6] for fault classification. The method used in [7] uses rough-cloud theory to tackle the problem of abundance and uncertainty of PQ data. Furthermore, a comparison of deterministic and statistical fault classification techniques is presented in [8].

On the contrary to the above discussion, not much work has been done in waveform characterization of fault events, which is the first step towards root-cause analysis. The goal of this paper is to utilize the waveform characteristics of fault events to distinguish between different fault types. The approach taken is to identify, extract and analyze characteristics of fault events based on their waveforms. These characteristic can be common to many faults types or unique (differentiating) to a particular fault type. A number of characteristics are explored in this paper, namely, time of occurrence of fault, number of affected phases, fault phase angle, time duration of fault, frequency content, and arc voltage. Some the characteristics like the number of affected phases and time duration of fault are common to all fault types. On the other hand, the characteristics like time of occurrence and arc voltage can be effectively employed to distinguish between fault types. The main contribution of this paper is that it provides a set of characteristics that can be extracted from any voltage and current waveforms obtained from a PQ monitor. A combination of these characteristics, rather than any one individual characteristic, can be used to uniquely identify the root-cause of the PQ events.

The introduction to the paper was provided in Section I. The PQ data used in this paper is described in Section II. Sections III to VIII describe the characteristic signatures based on, time stamp, fault insertion angle, number of phases, time duration, wavelet transforms and arc voltage, respectively. The conclusion is presented in section IX.

II. DATA SOURCE FOR POWER QUALITY DISTURBANCES

The data analyzed in this paper is obtained from PQ monitoring stations located in an electric supply utility in the Northeastern part of the United States. The utility has a network consisting of overhead power lines and also underground cables. The PQ data set has 143 tree contact cases, 66 animal contact cases and 28 lightning-induced cases. Lightning-induced cases refer to all the events that had lightning as the primary cause and not necessarily a direct lightning strike. The cause of the fault is identified by the utility after conducting a physical survey of the fault site. The cause of the fault is assumed as a fact throughout the paper. The PQ monitor waveforms at various voltage levels are

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sampled at the rate of 128 samples per cycle. Generally, the sampling window contains 10 cycles with the fundamental frequency of 60 Hz. The PQ data is analyzed based on the characteristic signatures explained in the following sections.

III. CHARACTERISTIC SIGNATURE: TIME STAMP

The first signature is based on the time stamp of the disturbance. Even though this is not an electrical quantity, it provides vital clues considering that power quality disturbances are influenced by the weather conditions seasonal and diurnal cycles. For example, lightning is associated with thunderstorms which generally occur in late spring and summer. Animal and bird contacts associated with power lines are more common during the migratory season. A similar analysis for animal contact events was presented in [9]. Based on the month of occurrence, events are classified into four types; Summer (June-August), Fall (September- November), Winter (December-February) and Spring (March -May). Fig. 1 shows this normalized seasonal classification.

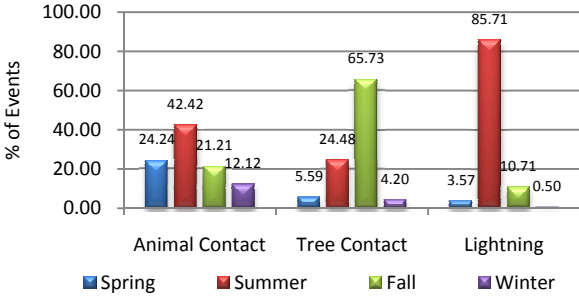


Fig. 1. Normalized Seasonal Distribution of Animal Contact, Tree Contact and Lightning Events

About 85% of lightning induced disturbances occur in summer, which happens to be the time of the year when thunderstorms are common. On the contrary, 65% tree contact events occur during fall when the wind is generally stronger and the trees are losing their leaves. Animal contact events are spread nearly evenly in spring and fall, with a majority (42.42%), occurring in summer.

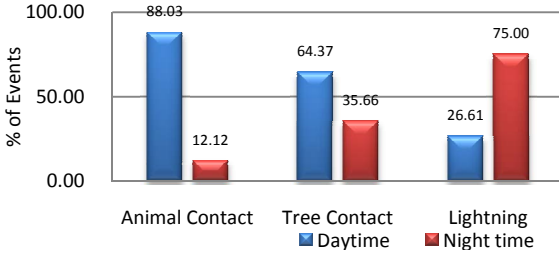


Fig. 2. Normalized Classification of Animal Contact, Tree Contact and Lightning Events based on time of day

Based on the time of day, events are classified as shown in Fig. 2. The disturbances between 6:00 am and 7:00 pm are categorized as 'daytime' events while 'night time' events are those that occur between 7:00 pm and 6:00 am. Nearly 90% of the animal contact triggered events occur during the daytime. This number is influenced by the kind of animals present in the geographical region of the utility. If a particular utility has

a large presence of nocturnal animals, the graph will be opposite to Fig. 2. 64% of the tree contact cases occur during the day, while three quarters of the lightning induced events occur after the sun has set.

IV. CHARACTERISTIC SIGNATURE: FAULT INSERTION ANGLE

Fault insertion angle can be defined as the angle of the normal sine wave at the disturbance initiation point. Initially, it was construed that the tree contact and animal contact triggered events would start around the peak of the sine wave as the voltage gradient between the animal/tree branch and the power line is highest at the peak. Fig. 3 and Fig. 4 show an animal contact and tree contact event, respectively, occurring near the positive peak of the sine wave. For the animal contact case the fault occurs in phase A, while in the tree contact case phase B is affected. The inception of the fault is characterized by a sudden sag in the voltage and a corresponding rise in the current of the same phase above the normal operating value.

However, it was found that some lightning induced events also occur at the peak of the sine wave as shown in Fig. 5. In this case the fault inception is at the negative peak of the voltage waveform and phase C is affected.

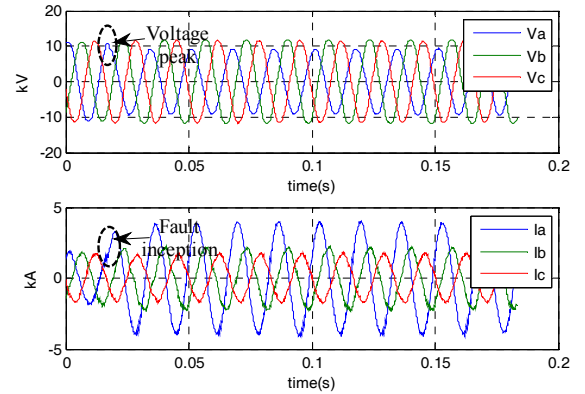


Fig. 3. Animal Contact Event Occurring at Voltage Peak

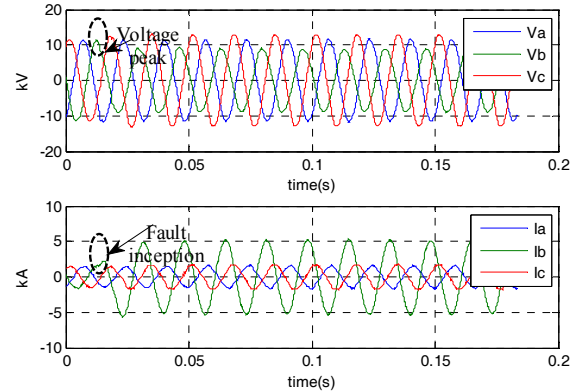


Fig. 4. Tree Contact Event Occurring at Voltage Peak

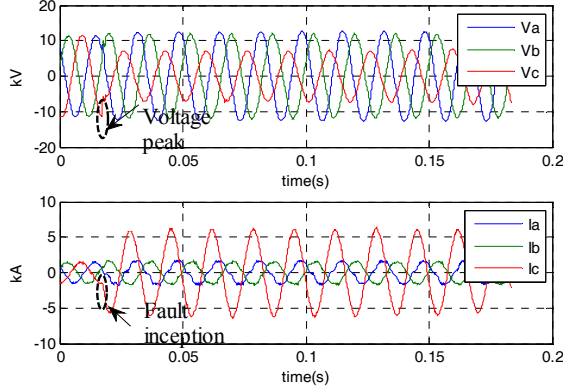


Fig. 5. Lightning Induced Event Occurring at Voltage Peak

The fault insertion angle results for a total of 170 cases are summarized in Table I, while 67 cases were found to be ambiguous. Ambiguous cases are mostly those in which the waveforms obtained from PQ monitoring station do not contain the required prefault data. If the fault insertion angle is between 15° of the positive or negative peak ($\pm 90^\circ$) of the sine wave, the event is said to have an inception at the peak. All other events have an off-peak fault insertion angle.

TABLE I
FAULT INSERTION ANGLE

Event	Insertion angle around peak	Off-peak Insertion angle	Total cases
Animal Contact	43 (70.5%)	18 (29.5%)	61
Tree Contact	66 (73.33%)	24 (26.67%)	90
Lightning Induced	16 (84.21%)	3 (15.79%)	19

About 70% of the animal and tree contact events have the fault insertion angle near the peak. However, it is observed that roughly 85% of lightning induced events also started near the voltage peak. This is because the cases analyzed in this paper are lightning-induced faults as opposed to direct lightning stroke events which can occur at any phase angle. Hence, the breakdown point for lightning-induced events, unlike direct strikes, is also the voltage peak. Thus, being initiated around the peak of a sinusoidal wave, with the available data set, does not seem to be a very unique feature of any particular group.

V. CHARACTERISTIC SIGNATURE: NUMBER OF PHASES

A fault can affect one, two, or all three phases of a distribution feeder. The number of phases involved in the fault depends on the cause of the fault. Generally, animal and tree contacts tend to affect only one phase and hence these events are single line to ground faults. On the contrary, lightning has the likelihood of affecting all three phases of the power line. The number of faulted phases is found out for all the 170 cases and the results are given in Table II. A disturbance affecting only one phase is called a single phase fault, while if two or three phases are affected the event is categorized as a multi-phase fault.

TABLE II
NUMBER OF AFFECTED PHASES

Event	Single phase faults	Multiphase faults	Total cases
Animal Contact	53 (86.88%)	8 (13.12%)	61
Tree Contact	66 (73.33%)	24 (26.67%)	90
Lightning Induced	8 (42.10%)	11 (57.90%)	19

As many as 53 out of 61 animal contact events and 66 out of 90 tree contact events, i.e. more than 70%, are single phase faults. On the other hand, roughly 60% lightning induced events are multiphase. As in the case of the fault insertion angle, the number of phases involved is not a characteristic signature of any particular event. Further research needs to be conducted in order to better utilize this feature.

VI. CHARACTERISTIC SIGNATURE: TIME DURATION

Characteristically lightning induced events are high-frequency transients having higher energy content as compared to contact triggered events. The AC resistance and line impedance of an overhead conductor is directly proportional to increasing frequency [10]. This increased opposition to the flow of higher frequency currents makes high frequency transients like lightning die down faster as compared to relatively lower frequency transients like those due to tree/animal contact triggered disturbances. However, this may not always be the case as seen in Fig. 5 where the lightning-induced fault in phase C lasts for several cycles. Fig. 6 shows a tree contact event in which the fault in phase A is temporary and has duration of just one cycle, because the tree branch initiating the fault burns and falls to the ground, before any protective device operates.

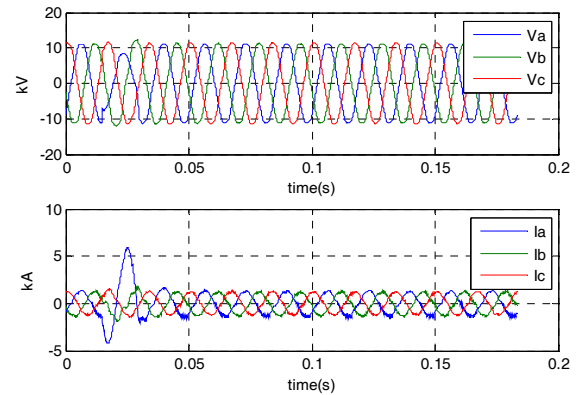


Fig. 6. Tree Contact Event Lasting for One Cycle

From the above examples it is clear that the time duration of the fault cannot be recognized as a unique characteristic of any particular event. The duration of the fault is determined by the operation of the overcurrent protective device interrupting the fault, rather than the cause itself.

VII. CHARACTERISTIC SIGNATURE: WAVELET TRANSFORMS

A lightning event can have one or more current pulses or strokes. The current stroke penetrates into the power system at the strike point and may be followed by one or more strokes.

The entire flash event occurs in a very short period, usually in few hundreds of microseconds. Induced voltage and current transients propagate away from the strike point and are captured by PQ monitors. Regardless of the location of PQ monitors, i.e., upstream or downstream from the lightning current injection, as long as they are located within the affected area, voltage and current waveforms should be captured almost simultaneously by all. Lightning induced waveforms may look quite different depending on the distance between the strike point and the PQ monitor. PQ monitors near the strike point are likely to capture high frequency oscillations at the initiation of the disturbance. While, those at different voltage levels or some electrical distance apart are unlikely to capture these high frequency oscillations, since overhead lines and transformers are inductive.

Fig. 7 and Fig. 8 show two sets of three-phase voltage and current waveforms caused by a lightning stroke on a 161-kV system at 10:01:22.000. Monitor T, downstream from strike, did not see the high current, while Monitor W-C, upstream from strike, saw it. Appreciable oscillatory transients (2.2 kHz) were present during the initiation of the disturbance. Fig. 9 shows a set of waveforms due to the same lightning stroke captured by a PQ monitor at a 13.8-kV system. The voltage and current waveforms are smooth and do not have an appreciable oscillatory transient.

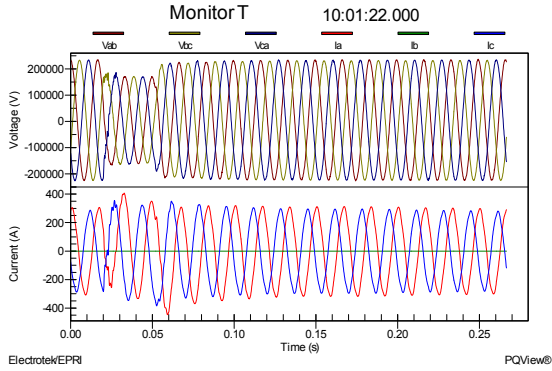


Fig. 7. Three-phase Voltage and Current Waveforms of Lightning Event Captured at 161-kV System. Monitor T Downstream from Lightning Stroke.

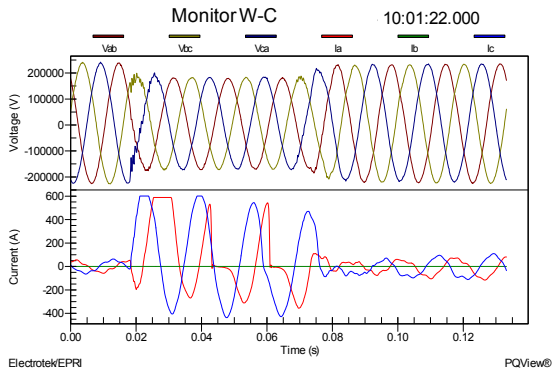


Fig. 8. Three-phase Voltage and Current Waveforms of Lightning Event Captured at 161-kV System. Monitor W-C Upstream from Lightning Stroke.

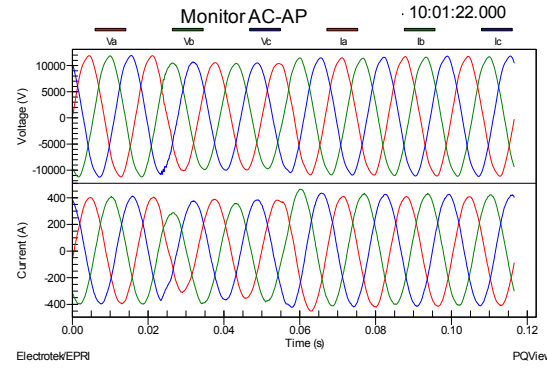


Fig. 9. Three-phase Voltage and Current Waveforms of Lightning Event Captured at 13.8 kV System by Monitor AC-AP

Wavelet transform can be used to analyze and uncover the spectral content in the time-scale domain. In this paper, orthogonal-dyadic wavelet transform is applied [11]. Only one phase is analyzed since results from other phases are similar. Voltage waveform is normalized and decomposed into three wavelet analysis scales; Scale 1(1920-3840 Hz), Scale 2(960-1920 Hz), Scale 3(480-960 Hz). They represent the time-scale energy spectra of the time domain signal.

The discrete wavelet transform of the waveform captured by Monitor W-C (Fig. 8) is shown in Fig. 10. The wavelet transform of the Monitor AC-AP is computed on similar lines.

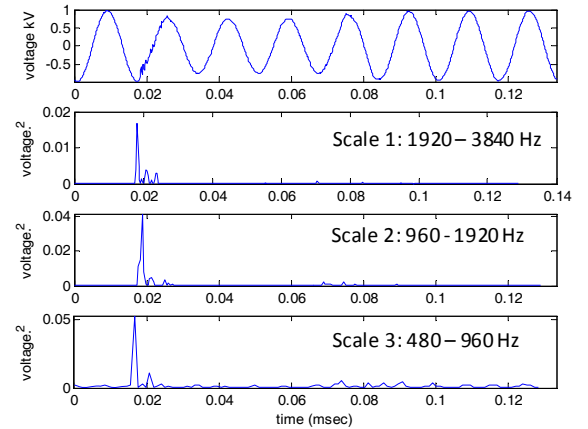


Fig. 10. A Three-scale Wavelet Transform of Lightning induced Voltage Waveform Captured by Monitor W-C (161 kV)

By analyzing the wavelet transforms it was clear that transients with high frequency contents appear in lower scales while those with lower frequency content appear in higher scales. Comparing the square of the transform outputs, on a scale by scale basis, for the voltage waveform captured by Monitor W-C, the maximum energy in scale 1, 2, and 3 is 1.7×10^{-2} , 4.3×10^{-2} , and 5.4×10^{-2} , respectively. In contrast, the voltage waveform captured by Monitor AC-AP has maximum energies in the first two scales as 2.6×10^{-3} and 7.8×10^{-4} , while there is no meaningful maximum energy information in the third scale. These results indicate that lightning induced waveforms may be identifiable in the wavelet domain if a PQ monitor is near the strike point. Lightning induced waveforms captured by remote PQ monitors can be identified indirectly through the use of time-proximity cross-correlation approach and other identifying signatures described in this paper.

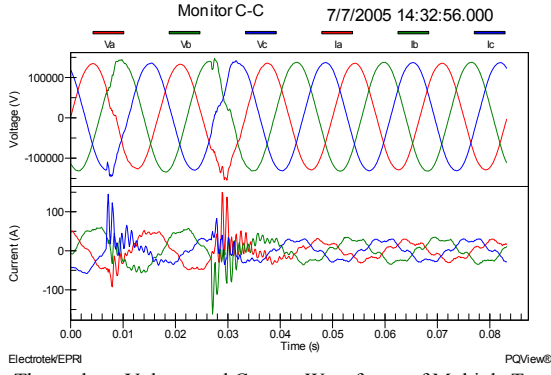


Fig. 11. Three-phase Voltage and Current Waveforms of Multiple Tree Contacts Captured at 161-kV System (Monitor C-C)

Tree contact events also have a number of wave shapes and are dependent on the voltage level where the tree contact occurs. Fig. 11 and Fig. 12 show simultaneous tree contacts captured by Monitor C-C (161 kV) and AC-AP (13.8 kV), respectively. Unlike tree contacts on distribution lines, they generally do not result in a permanent fault. Waveforms captured at transmission system level do possess more significant transients than those captured at a distribution system level. This is because inductive elements (transformers and overhead lines) of the power system offer higher impedance at a higher frequency. The square of the wavelet transform of voltage waveform captured by Monitor C-C is shown in Fig. 13. The frequency of the decomposition scale is the same as the one used for lightning-induced events.

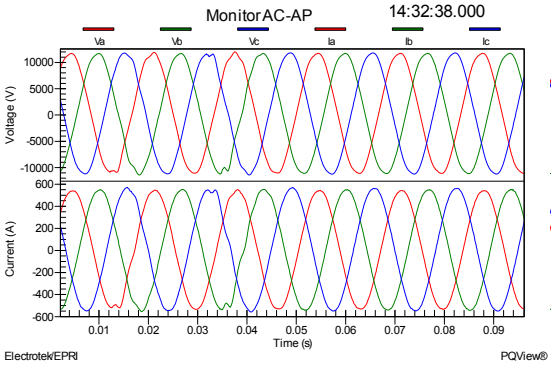


Fig. 12. Three-phase Voltage and Current Waveforms of Multiple Tree Contacts Captured at 13.8-kV System (Monitor AC-AP)

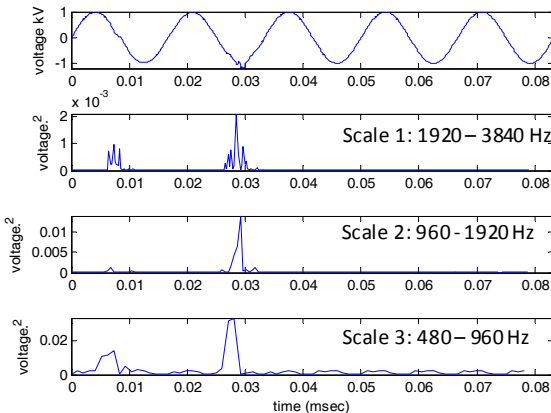


Fig. 13. Three-scale Wavelet Decomposition of Tree-contact Voltage Waveform Captured by Monitor C-C (161 kV)

The wavelet transforms for Monitor AC-AP is computed on similar lines. For the voltage waveform captured by Monitor C-C (161 kV), the maximum energy in scale 1, 2, and 3 is 2.1×10^{-3} , 1.39×10^{-2} , and 3.3×10^{-2} , respectively. In contrast, in the voltage waveform captured by Monitor AC-AP, the maximum energy in each scale is 9.4×10^{-7} , 1.28×10^{-5} , and 1.2×10^{-4} . Comparing these values to those shown earlier for a lightning event; scale 1 value for a voltage waveform (captured at 161 kV) caused by a lightning event is approximately 10 times higher than that for a tree-contact event (captured at 161 kV also). For scale 2 and 3 values, they are comparable. Using very limited waveform samples, it is observed that lightning events appear to possess higher frequency content than tree contact events. This is reasonable given the fact that lightning tends to have higher frequency due to the travelling wave phenomena arising from the discontinuity (from a piece of equipment to another) encountered during pulse propagation.

VIII. CHARACTERISTIC SIGNATURE: ARC VOLTAGE

When an object (tree or animal), is about to be in contact with the power lines, there may be an electric arc struck in the medium (generally air) between the object and the line [12]. The characteristics of the arc such as the arc length, arc impedance and arc voltage hold crucial information about the cause of the disturbance [13] [14]. Arguably, the arc voltage is the most important characteristic out of all the above and can be used as a signature of the power quality disturbance. Based on the algorithm proposed in [15] [16], the arc voltage is assumed to be a square wave in phase with the current. The algorithm is applicable to single phase faults and it estimates the arc voltage in the affected phase. It is assumed that the line impedance parameters are unknown. The single line diagram for a faulted circuit can be represented as shown in Fig. 14.

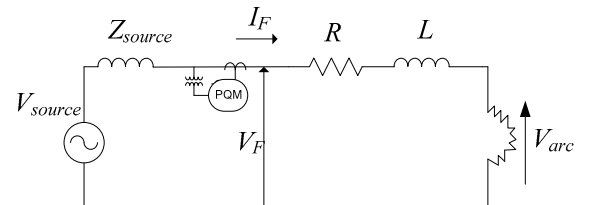


Fig. 14. Circuit for Arc Voltage Estimation

The voltage at the PQ monitoring site, V_F for the faulted phase can be written as shown in (1). This equation is applied at every point of the voltage and current waveform. The overdetermined system of equations is solved using least square method to obtain the peak value of the arc voltage, V_{arc} .

$$V_F = R \times I_F + L \left(\frac{dI_F}{dt} \right) + V_{arc} \times \text{sign}(I_F) \quad (1)$$

V_F is the fault phase voltage measured at PQ monitoring site
 I_F is the fault current

L is the line inductance

R is the line resistance

V_{arc} is the peak arc voltage at the fault location

$$\text{sign}(I_F) = 1, \text{ if } I_F > 0 \text{ and } -1 \text{ if } I_F \leq 0$$

Animal and tree contact triggered events involve physical contact of an object with power lines and hence tend to have an arc voltage higher than events induced by lightning strokes. Peak arc voltage estimates for animal contact (Fig. 3), tree contact (Fig. 4) and lightning induced events (Fig. 5) are shown in Fig. 15, Fig. 16 and Fig. 17, respectively.

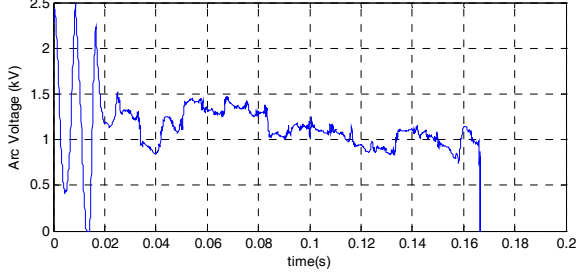


Fig. 15. Arc Voltage in Phase A for Animal Contact Event shown in Fig. 3

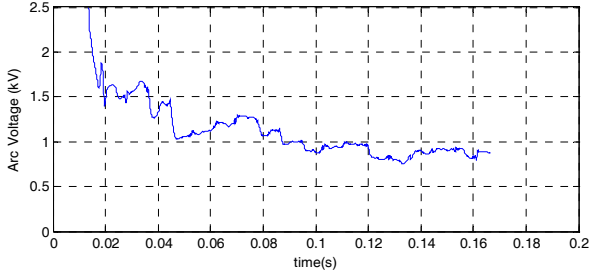


Fig. 16. Arc Voltage in Phase B for Tree Contact Event shown in Fig. 4

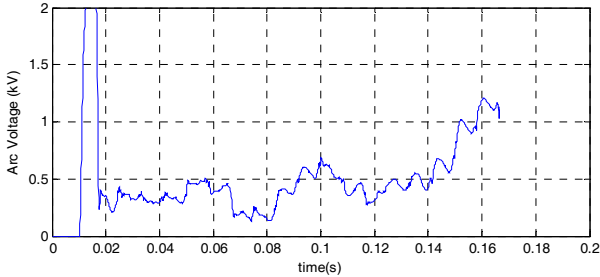


Fig. 17. Arc Voltage in Phase C for Lightning Induced Event shown in Fig. 5

The animal and tree contact events have a peak arc voltage that is more than 1 kV, while the lightning induced event has an arc voltage that remains largely below 1 kV for the fault duration. As mentioned earlier, the arc voltage estimation algorithm is applicable only to single phase faults. In addition, only the events that have a stable arc voltage value have been selected for analysis, resulting in a total of 20 animal contact cases, 24 tree contact cases and 14 lightning induced cases. The arc voltage values for these cases are plotted in a normalized fashion in Fig. 18. 25 cable failure cases from the same PQ data set are included for the purpose of comparison.

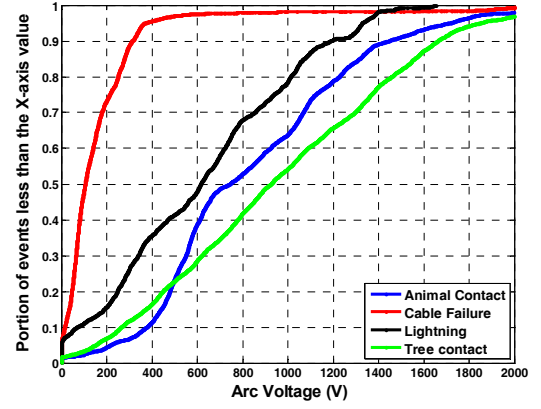


Fig. 18. Normalized Arc Voltage Distribution for Animal Contact, Tree Contact and Lightning Events

It is observed that lightning induced events have the least arc voltage among the three categories, while tree contact events have the highest arc voltage values. For the most part, animal contact events have an arc voltage in between lightning induced and tree contact events. Compared with these events, cable failure disturbances have much less arc voltage values.

IX. CONCLUSION

The information present in the waveforms captured by PQ monitoring stations was extracted in form of characteristic signatures which can be used for event classification. The signatures analyzed were the time of occurrence, fault insertion angle, number of affected phases, time duration of fault, wavelet domain analysis and arc voltage. Using the time stamp, the events were classified according to the season and also the time of day. Majority of the lightning induced events occurred in summer while, most of the tree contact events took place in fall. Both the animal and tree contact events were generally affecting only one phase of the distribution line, and these events usually had the fault insertion angle near the peak of the sine wave. The time duration of the fault proved to be an inconclusive characteristic. It was suggested that lightning events could be distinguishable from tree contact events in the wavelet domain. Furthermore, lightning induced events had an arc voltage that was lower than both the contact events.

Through the course of this analysis it was found that no single signature can exclusively identify a given event. However, a combination of these signatures can be used to uniquely characterize the animal contact, tree contact and lightning induced events. Moreover, several other characteristics like p.u. voltage drop in the faulted phase can be extracted in the future to determine the root-cause of faults. Future work involves quantification of the relevance of these characteristics and extraction of patterns to form a rule set. Such a set of rules can be used to identify causes of unknown events. The development of a rule based classification methodology for animal contact, tree contact and lightning induced events is tackled in [17].

X. ACKNOWLEDGEMENT

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XII. BIOGRAPHIES

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